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United States Patent Application Publication

20250277978

Kind Code

A1

Publication Date

September 04, 2025

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PROJECTION OPTICAL SYSTEM AND GLASSES-TYPE TERMINAL

Abstract

A projection optical system including: a projection substrate for projecting an image light onto a second surface while transmitting at least a part of light, that is incident from a first surface, to the second surface opposite to the first surface, the projection substrate having an optical waveguide; and a polarization reduction plate that is provided on the first surface side with an air layer between itself and the optical waveguide, covers at least a part of the optical waveguide, and reduces light with a polarization direction from the image light leaking from the first surface of the projection substrate, wherein the optical waveguide guides at least a part of projection light for projecting the image light, and emits the projection light as the image light from the second surface.

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Appl. No.: 19/046535

Filed: February 06, 2025

Related U.S. Application Data

parent WO continuation PCT/JP2022/030272 20220808 PENDING child US 19046535

Publication Classification

Int. Cl.: G02B27/01 (20060101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation application of International Application number PCT/JP2022/030272, filed on Aug. 8, 2022, contents of which are incorporated herein by reference in their entirety. [0002] The present invention relates to a projection optical system and a glasses-type terminal.

BACKGROUND OF THE INVENTION

[0003] Conventionally, an eyeglass-type device, a head mounted display, and the like have been known for displaying two-dimensional images to be observed by a user, utilizing an optical system including a waveguide and the like (for example, refer to Japanese Unexamined Patent Publication No. 2017-207686).

[0004] Since such devices incorporate the optical system into limited space, the optical system may become complex. Additionally, a part of the image light projected for the user's observation may be emitted as leakage light in a direction other than toward the user, potentially making the user's eyes appear to glow.

BRIEF SUMMARY OF THE INVENTION

[0005] In view of these issues, the present invention aims to reduce leakage light of image light intended for the user's observation with a simple configuration.

[0006] A first aspect of the present invention provides a projection optical system including: a projection substrate for projecting an image light onto a second surface while transmitting at least a part of light, that is incident from a first surface, to the second surface opposite to the first surface, the projection substrate having an optical waveguide; and a polarization reduction plate that (i) is provided on the first surface side with an air layer between itself and the optical waveguide, (ii) covers at least a part of the optical waveguide, and (iii) reduces light with a polarization direction from the image light leaking from the first surface of the projection substrate, wherein the optical waveguide guides at least a part of projection light for projecting the image light, and emits the projection light as the image light from the second surface.

[0007] A second aspect of the present invention provides a glasses-type terminal worn by a user, including: the projection optical system according to claim 1, which is provided as at least one of a lens for the right eye and a lens for the left eye of the user, and projects the image light onto the second surface while transmitting at least a part of light incident from the first surface to the eye of the user; a frame that fixes the projection substrate; and a projection unit that is provided in the frame and radiates the projection light, for causing the image light to be projected onto an emission region of the optical waveguide, onto an incident region of the optical waveguide of the projection substrate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 shows a first configuration example of a glasses-type terminal 10 according to the present embodiment.

[0009] FIG. 2 shows an outline of an optical path of a projection light in the glasses-type terminal 10 according to the present embodiment.

[0010] FIG. 3 shows an outline of the optical path of the projection light in a projection substrate

100 according to the present embodiment.

[0011] FIG. **4** shows an example of a projection light which is radiated from a projection unit **120** according to the present embodiment to the projection substrate **100**, and an image light emitted from the projection substrate **100**.

[0012] FIG. **5** shows a configuration example of the projection substrate **100** according to the present embodiment.

[0013] FIG. **6** shows a second configuration example of the glasses-type terminal **10** according to the present embodiment.

[0014] FIG. **7** shows a third configuration example of the glasses-type terminal **10** according to the present embodiment.

[0015] FIG. **8** shows a fourth configuration example of the glasses-type terminal **10** according to the present embodiment.

DETAILED DESCRIPTION OF THE INVENTION

[0016] Hereinafter, the present invention will be described through exemplary embodiments, but the following exemplary embodiments do not limit the invention according to the claims, and not all of the combinations of features described in the exemplary embodiments are necessarily essential to the solution means of the invention.

First Configuration Example of Glasses-Type Terminal **10**

[0017] FIG. **1** shows a first configuration example of a glasses-type terminal **10** according to the present embodiment. In this embodiment, three mutually orthogonal axes are designated as the X-axis, Y-axis, and Z-axis. The glasses-type terminal **10** is a wearable device worn by a user, for example. The glasses-type terminal **10** projects an image light onto a display region provided on a projection substrate **100** while having a user observe a view through glasses. The glasses-type terminal **10** includes a projection optical system **50**, a frame **110**, and a projection unit **120**.

[0018] The projection optical system **50** includes the projection substrate **100** and a polarization reduction plate **310**. In FIG. **1**, the projection substrate **100** of the projection optical system **50** is shown, and the polarization reduction plate **310** is omitted. The polarization reduction plate **310** will be described later.

[0019] The projection substrate **100** includes an optical waveguide **200**, and projects the image light onto a second surface while transmitting at least a part of the light entering from a first surface to the eyes of the user. Here, the first surface of the projection substrate **100** is a surface facing the side opposite to the user when the user is wearing the glasses-type terminal **10**. The second surface of the projection substrate **100** is a surface facing the user when the user is wearing the glasses-type terminal **10**. FIG. **1** shows an example in which the first surface and the second surface of the projection substrate **100** are disposed approximately parallel to an XY plane.

[0020] The projection substrate **100** is, for example, a glass substrate on which the optical waveguide **200** is formed. The optical waveguide **200** guides at least a part of projection light for projecting the image light incident from the second surface of the projection substrate **100**, and emits that part of the projection light as the image light from the second surface. The projection substrate **100** will be described later.

[0021] The frame **110** fixes the projection optical system **50**. The frame **110** is provided with the projection optical system **50** as at least one of a lens for the right eye or a lens for the left eye of the user. FIG. **1** shows an example in which a projection optical system **50a** is provided as the lens for the right eye of the user on the frame **110**, and a projection optical system **50b** is provided as the lens for the left eye.

[0022] Alternatively, the frame **110** may be provided with one projection optical system **50** as the lens for the right eye or the lens for the left eye of the user. Further, the frame **110** may be provided with one projection optical system **50** as a lens for both eyes of the user. In this case, the frame **110** may have a goggle shape. The frame **110** has parts such as a temple, a strap, and the like so that the user can wear the glasses-type terminal **10**.

[0023] The projection unit **120** is provided in the frame **110** and radiates the projection light, for causing the image light to be projected onto the projection substrate **100**, toward the projection optical system **50**. The frame **110** is provided with one or a plurality of such projection units **120**. FIG. **1** shows an example in which (i) a projection unit **120a** for irradiating the projection optical system **50a** (projection substrate **100a**) with a projection light L1 and (ii) a projection unit **120b** for irradiating the projection optical system **50b** (projection substrate **100b**) with a projection light L2 are provided in the frame **110**.

[0024] The projection unit **120** may be provided at a portion of the frame **110** to which the projection optical system **50** is fixed, or may be provided in the temple or the like of the frame **110**. The projection unit **120** is preferably provided integrally with the frame **110**. For example, the projection unit **120** radiates a projection light including one wavelength onto the projection optical system **50**, allowing the user to observe a monochrome image. Further, the projection unit **120** may radiate the projection optical system **50** with a projection light including a plurality of wavelengths, allowing the user to observe an image including multiple colors.

[0025] Next, the projection optical system **50** will be described. First, the operation of the projection substrate **100** within the projection optical system **50** will be described, followed by a description of a polarization reduction plate **310**.

[0026] FIG. **2** shows an outline of an optical path of a projection light in the glasses-type terminal **10** according to the present embodiment. The projection unit **120** radiates the projection light onto an incident region **210** of the optical waveguide **200** of the projection substrate **100**. The incident region **210** guides the projection light into a substrate of the projection substrate **100**. Then, at least a part of the projection light guided in the substrate is emitted as an image light from the emission region **230** of the optical waveguide **200**. The incident region **210** and the emission region **230** will be described later.

[0027] FIG. **3** shows an outline of an optical path of a projection light in the projection substrate **100** according to the present embodiment. As will be described later, the optical waveguide **200** includes the incident region **210**, an intermediate region **220**, and the emission region **230**. A projection light L enters the incident region **210** and is emitted from the emission region **230** through the intermediate region **220** as an image light P. The intermediate region **220** guides the projection light L to the emission region **230**, part by part, as the projection light L travels away from the incident region **210**.

[0028] Similarly, as the projection light L travels away from the intermediate region **220**, the emission region **230** also emits portions of the projection light L as part of the image light P. By doing this, the projection substrate **100** emits, as the image light P, the projection light L incident on the incident region **210** from the emission region **230**.

[0029] Here, an example is conceived of in which the intermediate region **220** guides the projection light L to the emission region **230** at a constant rate throughout the entire region of the intermediate region **220**. In this case, since the quantity of the projection light L decreases as the projection light L travels away from the incident region **210**, the intensity of the projection light L entering the emission region **230** from the intermediate region **220** may differ depending on a distance from the incident region **210**.

[0030] Similarly, an example is conceived of in which the emission region **230** emits, as the image light P, the projection light L at a constant rate throughout the entire region of the emission region **230**. In this case, since the quantity of the projection light L decreases as the projection light L travels away from the intermediate region **220**, the intensity of the image light P emitted from the emission region **230** may differ depending on a distance from the incident region **210** and a distance from the emission region **230**. For example, luminance may gradually decrease from the upper left pixels to the lower right pixels of an image projected by the emission region **230**. The projection substrate **100** according to the present embodiment reduces such variations in the luminance.

Example of the Projection Light and Image Light

[0031] FIG. 4 shows an example of the projection light L radiated from the projection unit **120** to the projection substrate **100** and the image light P emitted from the projection substrate **100** according to the present embodiment. For example, the projection unit **120** radiates the projection light L toward the second surface of the projection substrate **100** positioned in the Z direction. The projection light L corresponds to an image to be shown to the user, and for example, when a screen or the like is installed on a plane approximately parallel to the XY plane and the projection light L is projected thereon, an image M1 to be observed by the user is displayed on that screen. The image to be shown to the user is an AR (Augmented Reality) image or a VR (Virtual Reality) image generated by a processor included in the projection unit **120**, for example. In this way, the projection unit **120** radiates, as the projection light L, a plurality of light rays forming the image M1 on the plane approximately parallel to the XY plane.

[0032] In this embodiment, an example in which the projection unit **120** projects an approximately rectangular image M1, whose longitudinal direction is the X-axis direction on the plane, approximately parallel to the XY plane will be described. In FIG. 4, five light rays, from among the plurality of light rays radiated by the projection unit **120**, are shown as input light rays **20**. For example, a light ray corresponding to the upper left pixels of the image is a first input light ray **20a**, a light ray corresponding to the lower left pixels of the image is a second input light ray **20b**, a light ray corresponding to the center pixels of the image is a third input light ray **20c**, a light ray corresponding to the upper right pixels of the image is a fourth input light ray **20d**, and a light ray corresponding to the lower right pixels of the image is a fifth input light ray **20e**.

[0033] For example, the projection unit **120** irradiates the incident region **210** of the projection substrate **100** with such projection light L so as to form an upright virtual image at infinity or at a predetermined position. The projection light incident on the incident region **210** passes through the intermediate region **220** and is emitted from the emission region **230** as the image light P. The image light P is emitted from the emission region **230** and enters the user's eyes, which are at a distance d from the projection substrate **100**. The image light P forms an image M2 on the retina of the user's eyes. In this way, the image light P includes a plurality of light fluxes that form the image M2.

[0034] In FIG. 4, five light fluxes, from among a plurality of light fluxes which are radiated from a circular region C of the emission region **230** of the projection substrate **100** and formed into an image at a predetermined position, are shown as output light fluxes **30**. For example, a light flux formed into an image as the lower right pixels of the image M2 is designated as a first output light flux **30a**, a light flux formed into an image as the upper right pixels of the image M2 is designated as a second output light flux **30b**, a light flux formed into an image as the center pixels of the image M2 is designated as a third output light flux **30c**, a light flux formed into an image as the lower left pixels of the image M2 is designated as a fourth output light flux **30d**, and a light flux formed into an image as the upper left pixels of the image M2 is designated as a fifth output light flux **30e**.

[0035] Each light flux corresponds to one of the plurality of input light rays **20** entering from the projection unit **120**. For example, the first output light flux **30a** corresponds to the first input light ray **20a**, and the first output light flux **30a** includes a plurality of light rays generated by a plurality of splittings, diffractions, and the like of the first input light ray **20a** that take place from the incident region **210** to the emission region **230** of the projection substrate **100**. Similarly, the second output light flux **30b** corresponds to the second input light ray **20b**, the third output light flux **30c** corresponds to the third input light ray **20c**, the fourth output light flux **30d** corresponds to the fourth input light ray **20d**, and the fifth output light flux **30e** corresponds to the fifth input light ray **20e**.

[0036] In other words, the image M2, which is the image light P emitted from the emission region **230** and formed on the retina of the user's eyes, corresponds to the image M1 projected with the projection light L radiated by the projection unit **120**. In this way, the user wearing the glasses-type

terminal **10** can perceive the image M2 as if it were projected onto the second surface of the projection substrate **100**, superimposed on a view seen through the projection substrate **100**. In other words, the emission region **230** functions as the display region for displaying the image M2 corresponding to the image M1 projected with the projection light L.

[0037] In FIG. 4, the image M2 observed by the user is an image obtained by inverting the image M1 projected with the projection light L vertically and horizontally. The image M1 projected with the projection light L may be a still image, or instead, may be a moving image. The projection substrate **100** that emits the image light P corresponding to the incident projection light L will now be described.

Configuration Example of the Projection Substrate **100**

[0038] FIG. 5 shows a configuration example of the projection substrate **100** according to the present embodiment. FIG. 5 shows an example in which the first surface and the second surface of the projection substrate **100** are disposed approximately parallel to the XY plane. The projection substrate **100** is a substrate including the optical waveguide **200** for projecting the image light onto the second surface, which is the opposite side of the first surface, while transmitting at least a part of the light incident from the first surface to the second surface. The projection substrate **100** is a glass substrate, for example. The projection substrate **100** includes the optical waveguide **200** that includes the incident region **210**, the intermediate region **220**, and the emission region **230**.

Example of the Incident Region **210**

[0039] A projection light for projecting an image light enters the incident region **210**, and the incident region **210** guides the incident projection light toward the intermediate region **220**. FIG. 5 shows an example in which the incident region **210** has a circular shape in a plane approximately parallel to the XY plane, but the present invention is not limited thereto. The incident region **210** may have a shape such as an ellipse, a polygon, or a trapezoid, as long as it can guide the projection light to the intermediate region **220**.

[0040] The incident region **210** includes an input diffraction grating in which a plurality of first grooves **212** are formed with a first period. In other words, the plurality of first grooves **212** are arranged on the upper surface of the projection substrate **100** in the same direction with a predetermined groove width and interval, thereby functioning as the diffraction grating. The incident region **210** has a reflective or transmissive input diffraction grating and guides the projection light in a direction of the intermediate region **220** through reflective or transmissive diffraction. The first period of the plurality of first grooves **212** is in a range of about 10 nm to about 10 μm , for example.

[0041] The plurality of first grooves **212** are arranged in a direction from the incident region **210** toward the intermediate region **220**, for example. Here, the traveling direction of the projection light from the incident region **210** toward the intermediate region **220** is referred to as a first direction. FIG. 5 shows an example in which the first direction is a direction approximately parallel to the X-axis direction, and the first grooves **212** extending in a direction approximately parallel to the Y-axis direction are arranged in the first direction. Since the projection light converges as it enters the incident region **210**, the incident region **210** guides the projection light to the intermediate region **220** such that the projection light spreads out at a divergence angle centered on the first direction within the plane of the projection substrate **100**.

Example of the Intermediate Region **220**

[0042] The intermediate region **220** guides a part of the projection light incident from the incident region **210** toward the emission region **230**. The intermediate region **220** is provided in a region through which the projection light passes, in the plane approximately parallel to the XY plane. The intermediate region **220** has a reflective intermediate diffraction grating, and guides the projection light toward the emission region **230** through the reflective diffraction. The intermediate region **220** has a rectangular shape whose longitudinal direction is the first direction, for example.

[0043] Since the projection light travels while spreading out around the first direction, it is

preferable for the intermediate region **220** to have a shape that widens as the distance from the incident region **210** increases, diverging from the first direction, which is a traveling direction of the projection light passing through the incident region **210**. The intermediate region **220** has a trapezoidal shape, a fan shape, or the like in the plane approximately parallel to the XY plane, for example. FIG. 5 shows an example in which the intermediate region **220** has the trapezoidal shape. An intermediate region **220** with such a shape can be formed to correspond to a region where the projection light spreads while travelling in the XY plane, and can efficiently guide the projection light.

[0044] The intermediate region **220** includes an intermediate diffraction grating in which a plurality of second grooves **222** are formed with a second period. In other words, the plurality of second grooves **222** are arranged on the upper surface of the projection substrate **100** in the same direction with a predetermined groove width and interval, thereby functioning as the diffraction grating. The intermediate region **220** functions as, for example, a reflective intermediate diffraction grating, and guides the projection light to the emission region **230**.

[0045] The second period of the plurality of second grooves **222** is different from the first period of the plurality of first grooves **212**. As the second period, it is desirable to select an appropriate period for guiding the projection light to the emission region **230**. The second period is, for example, in a range of about 10 nm to about 10 μm .

[0046] The plurality of second grooves **222** are arranged in a predetermined direction, for example. For example, a direction from the intermediate region **220** toward the emission region **230** is defined as a second direction, and an angle formed between the first direction and the second direction is defined as a first angle. In this case, the plurality of second grooves **222** are formed in a direction inclined toward the second direction by an angle of $\frac{1}{2}$ of the first angle with respect to the first direction. FIG. 5 shows an example in which the second direction is a direction approximately parallel to the Y-axis direction, the first angle is approximately 90 degrees, and the plurality of second grooves **222** are arranged in the direction inclined toward the second direction by approximately 45 degrees with respect to the first direction.

[0047] The intermediate region **220** includes a plurality of first divided regions **224** arranged in the traveling direction of the incident projection light. The second grooves **222** formed in the plurality of first divided regions **224** have different depths. In other words, in the intermediate region **220**, the second grooves **222** are formed such that a ratio of light guided to the emission region **230** within the incident projection light varies for each of the first divided regions **224**.

[0048] The intermediate region **220** preferably includes three or more first divided regions **224**. In this way, the intermediate region **220** is divided into the plurality of first divided regions **224**, thereby varying the quantity of projection light guided to the emission region **230** for each of the first divided regions **224**. By doing this, the distribution of the quantity of light in a direction perpendicular to the traveling direction of the projection light is adjusted to be approximately constant, while guiding the projection light with different intensities, depending on the distance from the incident region **210**, to the emission region **230**.

[0049] For example, the second grooves **222** are formed in such a way that the depth of the second groove **222** provided in one of the first divided regions **224** is greater than the depth of the second groove **222** provided in the first divided region **224**, which is closer to the incident region **210** than that particular divided region **224**. In this case, the rate of change of depth of the second grooves **222** of two adjacent first divided regions **224** among the plurality of first divided regions **224** may increase as the distance from the incident region **210** increases.

[0050] As an example, an intermediate region **220** having three first divided regions **224**, as shown in FIG. 5, is considered. Here, it is assumed that a second groove **222a** is formed with a depth such that the second groove **222a** guides light with approximately $\frac{1}{4}$ of the quantity of the projection light incident on a first divided region **224a** to the emission region **230** in the first divided region **224a**, which is closest to the incident region **210** among the three first divided regions **224**. In this

case, approximately $\frac{3}{4}$ of the remaining quantity of the projection light incident on the first divided region **224a**, which is closest to the incident region **210**, enters an adjacent first divided region **224b**.

[0051] It is assumed that a second groove **222b** is formed with a depth such that the second groove **222b** guides light with approximately $\frac{1}{3}$ of the quantity of the projection light incident on the first divided region **224b** to the emission region **230** in the first divided region **224b**, which is second closest to the first divided region **224**. In other words, the depth of the second groove **222b** of the first divided region **224b**, which is second closest to the incident region **210**, is greater than the depth of the second groove **222a**, so as to guide light having $\frac{4}{3}$ times the quantity of light compared to the first divided region **224a**, which is closest to the incident region **210**, to the emission region **230**. The first divided region **224b** guides light with approximately $\frac{1}{4}$ of the quantity of the projection light incident on the first divided region **224a**, which is closest to the incident region **210**, to the emission region **230**.

[0052] Then, approximately $\frac{1}{2}$ of the remaining quantity of the projection light incident on the first divided region **224a**, which is closest to the incident region **210**, enters an adjacent first divided region **224c**. It is assumed that a second groove **222c** is formed with a depth such that the second groove **222c** guides light with approximately $\frac{1}{2}$ of the quantity of the projection light incident on the first divided region **224c** to the emission region **230** in the first divided region **224c**, which is third closest to the incident region **210**. In other words, the depth of the second groove **222c** of the first divided region **224c**, which is third closest to the incident region **210**, is greater than the depth of the second groove **222b**, so as to guide light having $\frac{3}{2}$ times the quantity of light compared to the first divided region **224b**, which is closest to the incident region **210**, to the emission region **230**.

[0053] In addition, the second grooves **222** of the two adjacent first divided regions **224** among the three first divided regions **224** are formed in such a way that the rate of change of depth of these second grooves **222** increases as the distance from the incident region **210** increases. Then, the first divided region **224c**, which is third closest to the incident region **210**, guides light with approximately $\frac{1}{4}$ of the quantity of the projection light incident on the first divided region **224a**, which is closest to the incident region **210**, to the emission region **230**. As in the above example, it can be understood that by adjusting the quantity of the projection light to be guided to the corresponding emission region **230** for each of the first divided regions **224** to a predetermined value, the intermediate region **220** can guide the projection light to the emission region **230** while ensuring approximately constant distribution of the quantity of the projection light guided to the respective emission regions **230**.

[0054] The intermediate region **220** may further include a first reflection region **226** at a position furthest from the incident region **210**. FIG. 5 shows an example in which the intermediate region **220** includes three first divided regions **224** and the first reflection region **226**. The first reflection region **226** reflects at least a part of the light that has passed through the plurality of first divided regions **224** to the plurality of first divided regions **224** again. The first reflection region **226** includes second grooves **222** of greater depth than the depth of the second grooves **222** of the adjacent first divided region **224**.

[0055] Since the intermediate region **220** includes such a first reflection region **226**, the plurality of first divided regions **224** guide at least a part of the light reflected by the first reflection region **226** to the emission region **230**. In this way, the intermediate region **220** can guide more projection light to the emission region **230**. The depth of the second grooves **222** of the plurality of first divided regions **224** may be determined such that the quantity of projection light guided to the emission region **230** from each of the first divided regions **224**, incorporating the reflected light from the first reflection region **226**, is approximately constant.

Example of the Emission Region **230**

[0056] The emission region **230** guides at least a part of the projection light incident from the

intermediate region **220** and emits that part of the projection light as an image light from the second surface of the projection substrate **100**. FIG. 5 shows an example in which the emission region **230** has a rectangular shape whose longitudinal direction is the X-axis direction in a plane approximately parallel to the XY plane, but the present invention is not limited thereto. The emission region **230** may have a rectangular shape, a square shape, a trapezoid shape, or the like whose longitudinal direction is the Y-axis direction, as long as the emission region **240** can guide the projection light and emit it as the image light.

[0057] The emission region **230** has an output diffraction grating in which a plurality of third grooves **232** are formed with a third period. In other words, the plurality of third grooves **232** are arranged on the upper surface of the projection substrate **100** in the same direction with a predetermined groove width and interval, thereby functioning as the diffraction grating. The emission region **230** has a reflective or transmissive output diffraction grating and guides the image light toward the user's eye through reflective or transmissive diffraction.

[0058] The third period of the plurality of third grooves **232** provided in the emission region **230** is different from the second period of the plurality of second grooves **222** in the intermediate region **220**. The third period of the plurality of third grooves **232** in the emission region **230** may be the same as the first period of the plurality of first grooves **212** in the incident region **210**. By making the period of the diffraction grating provided in the region into which the projection light enters approximately coincide with the period of the diffraction grating provided in the region where the image light is emitted in this manner, it is possible to reduce distortion or similar effects in an image observed by the user. The third period is in a range of about 10 nm to about 10 μ m, for example.

[0059] The plurality of third grooves **232** are arranged in the second direction, which is the direction from the intermediate region **220** toward the emission region **230**, for example. FIG. 5 shows an example in which the third grooves **232** extending in the first direction are arranged in the second direction.

[0060] Similarly to the intermediate region **220**, the emission region **230** includes a plurality of second divided regions **234** arranged in the traveling direction of the projection light incident from the intermediate region **220**. The third grooves **232** formed in the plurality of second divided regions **234** have different depths. In other words, in the emission region **230**, the third grooves **232** are formed such that a ratio of light which will be emitted as the image light within the incident projection light varies for each of the second divided regions **234**.

[0061] The emission region **230** preferably includes two or more second divided regions **234**. For example, the third groove **232** provided in one of the second divided regions **234** is assumed to have a depth greater than the depth of the third groove **232** provided in the second divided region **234**, which is closer to the intermediate region **220** than that particular second divided region **234**. Further, when the emission region **230** includes three or more second divided regions **234**, the rate of change of depth of the third grooves **232** of two adjacent second divided regions **234** may increase as the distance from the intermediate region **220** increases.

[0062] As described above, the emission region **230** is divided into the plurality of second divided regions **234**, resulting in variations in the quantity of light emitted as image light for each of the second divided regions **234**. In this way, similarly to the plurality of first divided regions **224** of the intermediate region **220**, by guiding the projection light as the image light, the emission region **230** can adjust the distribution of the quantity of light across the entire image to be approximately constant when observed by an observer as an image.

[0063] The emission region **230** may further include a second reflection region **236** at a position furthest from the intermediate region **220**. FIG. 5 shows an example in which the emission region **230** includes two second divided regions **234** and the second reflection region **236**. The second reflection region **236** reflects at least a part of the light that has passed through the plurality of second divided regions **234**, to the plurality of second divided regions **234** again. The second

reflection region **236** includes third grooves **232** of greater depth than the third grooves **232** of the adjacent second divided region **234**.

[0064] Since the emission region **230** includes such a second reflection region **236**, the plurality of second divided regions **234** emit, as the image light, at least a part of the light reflected by the second reflection region **236** from the second surface of the projection substrate **100**. In this way, the emission region **230** can emit more projection light as the image light, similarly to the intermediate region **220**. The depth of the third grooves **232** of the plurality of second divided regions **234** may be determined such that the quantity of light emitted as the image light from each of the second divided regions **234**, incorporating the reflected light from the second reflection region **236**, is approximately constant.

[0065] As described above, the projection substrate **100** according to the present embodiment splits the projection light entering the incident region **210** at different ratios for each of the plurality of first divided regions **224** of the intermediate region **220**, and emits them as image lights from the emission region **230**. By doing this, the projection substrate **100** can reduce variation in the luminance of the projection image to be observed by the user. In addition, the projection substrate **100** can further reduce variation in the luminance of the image by emitting the image light at different ratios for each of the plurality of second divided regions **234** in the emitting region **230**.

[0066] Such a projection substrate **100** can be realized by forming the diffraction grating corresponding to the incident region **210**, the diffraction grating corresponding the intermediate region **220**, and the diffraction grating corresponding the emission region **230** on the front or rear surface of the glass substrate or the like. The grooves forming the diffraction grating are made of resist, resin, or the like, for example. Therefore, the projection substrate **100** according to the present embodiment is a substrate that can be easily produced by forming grooves with predetermined intervals and depths for each region, without incorporating complicated optical systems.

Second Configuration Example of the Glasses-Type terminal **10**

[0067] Although an example of the glasses-type terminal **10** has been described above, in which one projection substrate **100** is provided on the frame **110** for each of the projection optical systems **50** for the right eye and the left eye, and the corresponding projection unit **120** irradiates the incident region **210** of each projection substrate **100** with the projection light, the present invention is not limited to this configuration. For example, one projection optical system **50** may include a plurality of projection substrates **100**. Next, the glasses-type terminal **10** configured in this manner will be described.

[0068] FIG. **6** shows a second configuration example of the glasses-type terminal **10** according to the present embodiment. In the glasses-type terminal **10** of the second configuration example, components that function in approximately the same manner as those of the glasses-type terminal **10** according to the present embodiment shown in FIG. **1** are denoted by the same reference numerals, and descriptions are omitted. The appearance of the glasses-type terminal **10** of the second configuration example may be approximately the same as that of the glasses-type terminal **10** shown in FIG. **1**.

[0069] A plurality of projection substrates **100** are fixed to the frame **110** of the glasses-type terminal **10** of the second configuration example. In this configuration, the plurality of projection substrates **100** are fixed to the frame **110** in such a way that emission regions **230**, provided on each of the plurality of projection substrates **100**, overlap at least partially in a planar view that is approximately parallel to the XY plane. FIG. **6** shows an example in which projection substrates **100R**, **100G**, and **100B** are fixed to the frame **110** of the glasses-type terminal **10**, and emission regions **230R**, **230G**, and **230B** of the three projection substrates **100** overlap each other in the planar view in the XY plane.

[0070] The projection unit **120** radiates projection lights of different wavelengths onto the corresponding incident regions **210** provided on each of the plurality of projection substrates **100**,

respectively. By doing this, the emission regions **230** provided on each of the plurality of projection substrates **100** respectively emit image light, corresponding to the projection lights respectively radiated onto the plurality of incident regions **210** from the projection unit **120**, from the second surface of the plurality of projection substrates **100** to the user's eyes.

[0071] Since the user wearing such a glasses-type terminal **10** observes an image in which the image lights of different wavelengths are superimposed, he/she can observe an image with colors resulting from color mixture. FIG. **6** shows an example in which the projection unit **120** radiates three projection lights corresponding to the three primary colors of RGB (such as red, green, and blue), which form an image, to the incident regions **210** of the three projection substrates **100**, respectively. Then, the three projection substrates **100** superimpose three image lights corresponding to the three primary colors of RGB and emit the superimposed lights to the user's eyes. By doing this, the user can observe an image having a plurality of colors of 2.sup.n, for example. Here, n is a positive integer such as 4, 8, 16, or 24.

Third Configuration Example of the Glasses-Type Terminal **10**

[0072] In the above glasses-type terminal **10**, a part of the image light that is supposed to be emitted toward the user may sometimes be emitted as leakage light in a direction other than that of the user. For example, a part of the image light emitting from the second surface of the projection substrate **100** may be unintentionally emitted from the first surface of the projection substrate **100** due to the diffraction grating of the optical waveguide **200**. In this case, a person looking at the user may perceive the user's eyes as glowing, which could cause discomfort. Therefore, the glasses-type terminal **10** according to the present embodiment may be configured to reduce such leakage light. This configuration will be described next.

[0073] FIG. **7** shows a third configuration example of the glasses-type terminal **10** according to the present embodiment. In the glasses-type terminal **10** of the third configuration example, components that function in approximately the same manner as those of the glasses-type terminal **10** according to the present embodiment shown in FIG. **1** are denoted by the same reference numerals, and descriptions are omitted. FIG. **7** is a diagram in which the projection unit **120** is omitted. The appearance of the glasses-type terminal **10** of the third configuration example may be approximately the same as that of the glasses-type terminal **10** shown in FIG. **1**.

[0074] In the third configuration example of the glasses-type terminal **10**, the projection optical system **50** further includes the polarization reduction plate **310**. The polarization reduction plate **310** is provided on the first surface side of the projection substrate **100**, with an air layer interposed between itself and the optical waveguide **200** of the projection substrate **100**. In this manner, the polarization reduction plate **310** is provided apart from the optical waveguide **200** so as not to influence optical characteristics of the optical waveguide **200**.

[0075] The polarization reduction plate **310** covers at least a part of the optical waveguide **200**, and reduces light with a polarization direction from image light leaking from the first surface of the projection substrate **100**. The polarization reduction plate **310** covers at least a part of the output diffraction grating of the emission region **230**. As a result, the polarization reduction plate **310** can receive the image light leaked from the output diffraction grating.

[0076] Since the image light leaked from the output diffraction grating is guided through a plurality of diffraction gratings of the optical waveguide **200**, the leaked image light becomes light polarized in one direction corresponding to the structure of the optical waveguide **200**. Therefore, the polarization reduction plate **310** is provided to reduce the light with the polarization direction from the image light leaked from the first surface of the projection substrate **100**.

[0077] As a result, the polarization reduction plate **310** can reduce the intensity of the leaked image light to a level where it is not noticeable to others when they look at the user wearing the glasses-type terminal **10**. Additionally, since the polarization reduction plate **310** transmits the light with a polarization direction perpendicular to that of the leaked image light, the polarization reduction plate **310** can transmit the external light, allowing the user to visually recognize it.

[0078] FIG. 7 illustrates an example in which the polarization reduction plate **310** includes a polarization filter provided facing the first surface of the projection substrate **100**. The polarizing filter is a polarizing plate, a polarizing film, or the like that attenuates the component of the incident light that is linearly polarized in a predetermined direction. The polarization reduction plate **310** is preferably fixed to the frame **110** or the projection substrate **100**. It should be noted that the polarization reduction plate **310** may be configured such that the polarization filter is rotatably provided, allowing adjustment of the polarization direction (absorption axis) of the light to be reduced. The polarization reduction plate **310** may be a polarizing film coated on a transparent substrate or the like. Next, the polarization reduction plate **310** configured in this manner will be described.

Fourth Configuration Example of the Glasses-Type Terminal **10**

[0079] FIG. 8 shows a fourth configuration example of the glasses-type terminal **10** according to the present embodiment. In the glasses-type terminal **10** of the fourth configuration example, components that function in approximately the same manner as those of the glasses-type terminal **10** according to the present embodiment shown in FIGS. 1 and 7 are denoted by the same reference numerals, and descriptions are omitted. The appearance of the glasses-type terminal **10** of the fourth configuration example may be approximately the same as that of the glasses-type terminal **10** shown in FIG. 1.

[0080] In the glasses-type terminal **10** of the fourth configuration example, the polarization reduction plate **310** includes a protective substrate **320** and a polarizing film **330**. The protective substrate **320** is provided facing the first surface of the projection substrate **100**. The protective substrate **320** is a substrate, such as a glass substrate or a plastic substrate that is transparent to at least visible light.

[0081] The polarizing film **330** is coated on at least one of a third surface or a fourth surface of the protective substrate **320**, the third surface being on the side opposite to the projection substrate **100** and the fourth surface facing the projection substrate **100**. FIG. 8 illustrates an example in which the polarizing film **330** is coated on the third surface of the protective substrate **320**.

[0082] Similar to the polarizing filter, the polarizing film **330** is a thin film that attenuates the component of the incident light that is linearly polarized in a predetermined direction. The polarizing film **330** may be coated on a part of or the entire protective substrate **320**.

[0083] Similar to the polarization reduction plate **310** described with reference to FIG. 7, the polarization reduction plate **310** having the protective substrate **320** and the polarizing film **330** can also reduce the intensity of the image light leaked from the output diffraction grating. The protective substrate **320** is preferably fixed to the frame **110** or the projection substrate **100**.

Additionally, the protective substrate **320** may be rotatably provided, and may be configured to allow adjustment of the direction of the absorption axis of the polarization reduction plate **310**.

[0084] The glasses-type terminal **10** according to the present embodiment described above can reduce the leakage light of the image light to be observed by the user, with a simple structure. It should be noted that the above-described example explains a case in which a transmissive axis of the polarization reduction plate **310** can be adjusted, but the present embodiment is not limited to this. Alternatively or additionally, the glasses-type terminal **10** may be configured such that the polarization direction of the image light can be adjusted.

[0085] For example, the projection unit **120** includes a polarization adjustment unit that adjusts a polarization direction of a projection light to be radiated onto the incident region **210**. The polarization adjustment unit includes, for example, a wave plate that rotates the polarization direction of linearly polarized light. The polarization adjustment unit adjusts the polarization direction of the image light leaking from the first surface of the projection substrate **100** so that it approximately aligns with the polarization direction in which the polarization reduction plate **310** reduces the intensity of the light. Such a projection unit **120** can adjust the polarization direction of the projection light so that leakage light is appropriately reduced by the polarization reduction plate

310 while efficiently guiding the projection light in the optical waveguide **200**.

[0086] Additionally, the polarization adjustment unit may adjust the polarization direction of the projection light so that the polarization direction of the image light leaked from the output diffraction grating becomes horizontal. In this case, the absorption axis of the polarization reduction plate **310** is approximately aligned with the horizontal direction. As a result, among the external light, horizontally polarized light is reduced by the polarization reduction plate **310**. Therefore, the glasses-type terminal **10** can also function as polarized glasses that reduce reflected light from horizontal surfaces such as water surfaces.

[0087] In the above-described third and the fourth configuration examples of the glasses-type terminal **10**, as described with reference to FIG. **6**, the projection optical system **50** may include the plurality of projection substrates **100** and allow the user to observe the image in which a plurality of image lights of different wavelengths are superimposed. In this case, it is preferable that the polarization directions of the plurality of image lights approximately align.

[0088] The polarization reduction plate **310** is preferably provided on the user-opposite side of the plurality of projection substrates **100**. FIG. **6** shows an example in which the projection optical system **50** includes three projection substrates **100** and one polarization reduction plate **310** provided on the user-opposite side of the three projection substrates **100**.

[0089] In this case, the projection unit **120** may include a polarization adjustment unit that adjusts the polarization direction of at least one projection light among a plurality of projection lights to be radiated onto the incident regions **210**. The polarization adjustment unit adjusts the polarization direction of at least one projection light from among a plurality of projection lights included in the image light leaking from the first surface of the projection substrate **100** so that it approximately aligns with the polarization direction in which the polarization reduction plate **310** reduces the intensity of the light. In this way, while efficiently guiding at least one projection light in the optical waveguide **200**, the projection unit **120** can adjust the polarization direction of the projection light so that leakage light corresponding to the projection light is appropriately reduced by the polarization reduction plate **310**.

[0090] It should be noted that the projection unit **120** may include a plurality of polarization adjustment units that adjust the polarization directions of a plurality of projection lights corresponding to the plurality of projection lights to be radiated onto the incident region **210**. In this case, the projection unit **120** can adjust the polarization directions of the plurality of projection lights so that leakage lights corresponding to the plurality of projection lights are appropriately reduced by the polarization reduction plate **310** while efficiently guiding the plurality of projection lights in the optical waveguide **200**.

[0091] In the glasses-type terminal **10** according to the present embodiment described above, an example has been described in which the optical waveguide **200** of the projection substrate **100** includes the incident region **210**, the intermediate region **220**, and the emission region **230**. However, the present invention is not limited to this configuration. The optical waveguide **200** only needs to be capable of outputting the projection light incident from the projection unit **120** as image light to be observed by the user, and the shapes or other aspects of the incident region **210**, intermediate region **220**, and emission region **230** are not limited to the described example. Additionally, the optical waveguide **200** may, for example, have a configuration in which it includes the incident region **210** and the emission region **230** but does not include the intermediate region **220**.

[0092] The present invention is explained based on the exemplary embodiments. The technical scope of the present invention is not limited to the scope explained in the above embodiments and it is possible to make various changes and modifications within the scope of the invention. For example, all or part of the apparatus can be configured with any unit which is functionally or physically dispersed or integrated. Further, new exemplary embodiments generated by arbitrary combinations of them are included in the exemplary embodiments. Further, effects of the new

exemplary embodiments brought by the combinations also have the effects of the original exemplary embodiments.

Claims

1. A projection optical system comprising: a projection substrate for projecting an image light onto a second surface while transmitting at least a part of light, that is incident from a first surface, to the second surface opposite to the first surface, the projection substrate having an optical waveguide; and a polarization reduction plate that (i) is provided on the first surface side with an air layer between itself and the optical waveguide, (ii) covers at least a part of the optical waveguide, and (iii) reduces light with a polarization direction from the image light leaking from the first surface of the projection substrate, wherein the optical waveguide guides at least a part of projection light for projecting the image light, and emits the projection light as the image light from the second surface.
2. The projection optical system according to claim 1, wherein the polarization reduction plate has: a polarization filter that is provided facing the first surface of the projection substrate.
3. The projection optical system according to claim 1, wherein the polarization reduction plate has: a protective substrate that is provided facing the first surface of the projection substrate; and a polarization film coated on at least one of a third surface or a fourth surface of the protective substrate, the third surface being on the side opposite to the projection substrate and the fourth surface facing the projection substrate.
4. The projection optical system according to claim 1, wherein the optical waveguide has an incident region, into which a projection light for projecting the image light enters, which includes an input diffraction grating and guides the incident projection light inside the projection substrate; and an emission region that includes an output diffraction grating, guides at least a part of the projection light incident from the incident region, and emits the at least part of the projection light as the image light from the second surface, and the polarization reduction plate covers at least a part of the output diffraction grating.
5. The projection optical system according to claim 4, wherein the optical waveguide further has an intermediate region that includes an intermediate diffraction grating and guides a part of the projection light incident from the incident region toward the emission region, in the input diffraction grating, a plurality of first grooves are formed with a first period, in the intermediate diffraction grating, a plurality of second grooves are formed with a second period, and in the output diffraction grating, a plurality of third grooves are formed with a third period.
6. A glasses-type terminal worn by a user, comprising: the projection optical system according to claim 1, which is provided as at least one of a lens for the right eye and a lens for the left eye of the user, and projects the image light onto the second surface while transmitting at least a part of light incident from the first surface to the eye of the user; a frame that fixes the projection substrate; and a projection unit that is provided in the frame and radiates the projection light, for causing the image light to be projected onto an emission region of the optical waveguide, onto an incident region of the optical waveguide of the projection substrate.
7. The glasses-type terminal according to claim 6, wherein the projection unit has a polarization adjustment unit that adjusts a polarization direction of the projection light to be radiated onto the incident region, and the polarization adjustment unit adjusts a polarization direction of the image light leaking from the first surface of the projection substrate and a polarization direction in which the polarization reduction plate reduces the intensity of light so as to match each other.
8. The glasses-type terminal according to claim 7, wherein the polarization adjustment unit adjusts the polarization direction of the projection light so that the polarization direction of the image light leaked from the optical waveguide becomes horizontal, and the polarization reduction plate matches the polarization direction, in which the polarization reduction plate reduces the intensity of light, to the horizontal direction.

9. The glasses-type terminal according to claim 6, wherein a plurality of the projection substrates are fixed to the frame, the polarization reduction plate is provided on the user-opposite side of a plurality of projection substrates, the projection unit radiates the projection light, including different wavelengths, onto the respective incident regions provided on each of the plurality of projection substrates, and the emission regions, provided on each of the plurality of projection substrates, overlap at least partially in a planar view, and respectively emit the image light, corresponding to the projection light radiated onto a plurality of the incident regions from the projection unit, from the second surface of the plurality of projection substrates to the eyes of the user.

10. The glasses-type terminal according to claim 9, wherein the projection unit has a polarization adjustment unit that adjusts a polarization direction of at least one projection light among a plurality of the projection lights to be radiated onto the incident regions, and the polarization adjustment unit adjusts a polarization direction of the image light corresponding to the at least one projection light, among a plurality of image lights leaking from the respective first surfaces of a plurality of the projection substrates, and a polarization direction in which the polarization reduction plate reduces the intensity of light so as to match each other.
